Dimension and Rank

Geometric Algorithms

CAS CS 132

Objectives

- Discuss the **coordinate systems**
- Introduce the fundamental notion of **dimension** which quantifies how "large" a space is
- Relate the dimension of the **column space** and the **null space** of a matrix (rank-nullity thm)

Keywords

basis

column space

null space

coordinate system

change of basis

dimension

rank

rank theorem

invertible matrix theorem (extended)

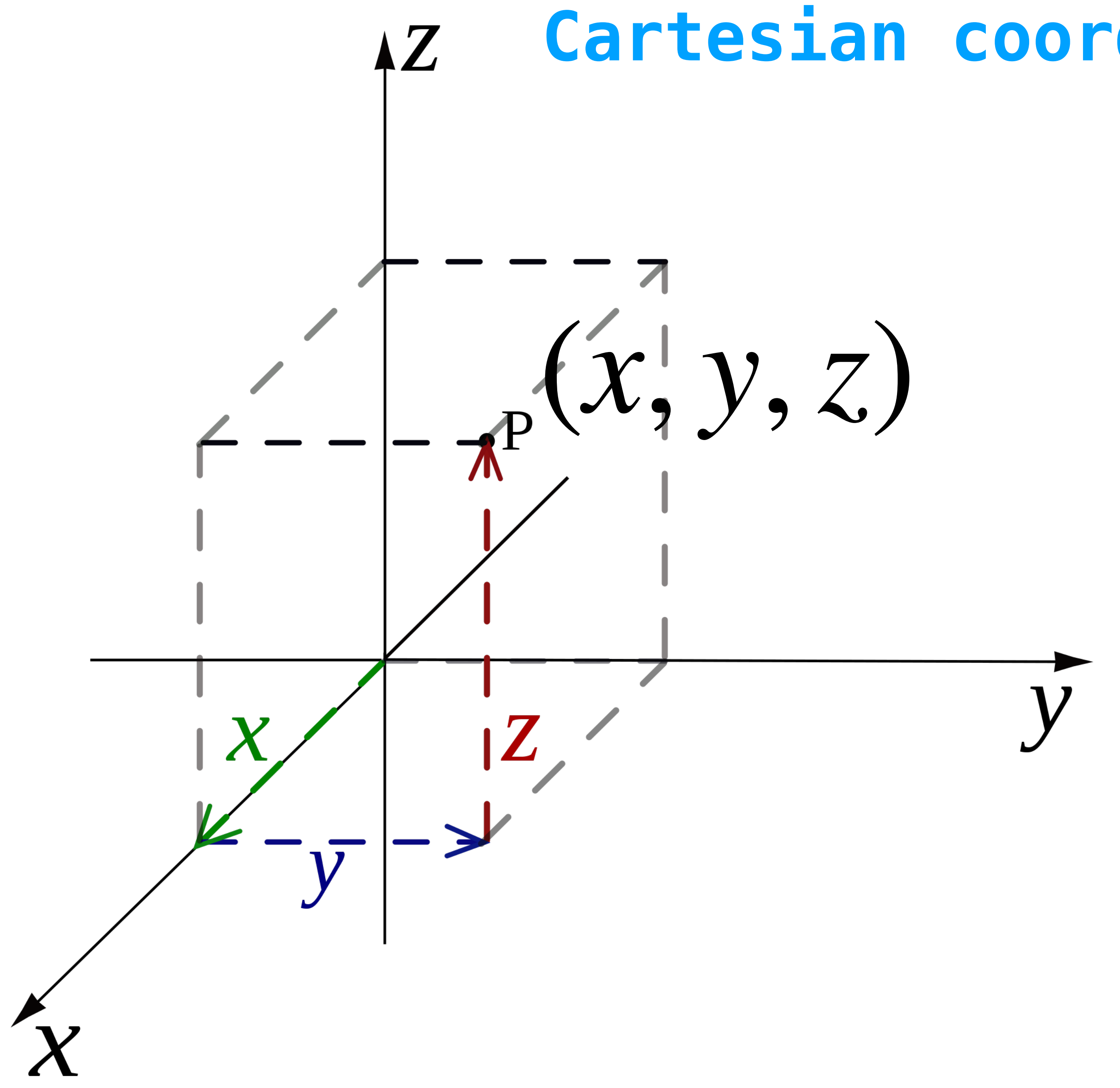
Coordinate Systems

At a High Level

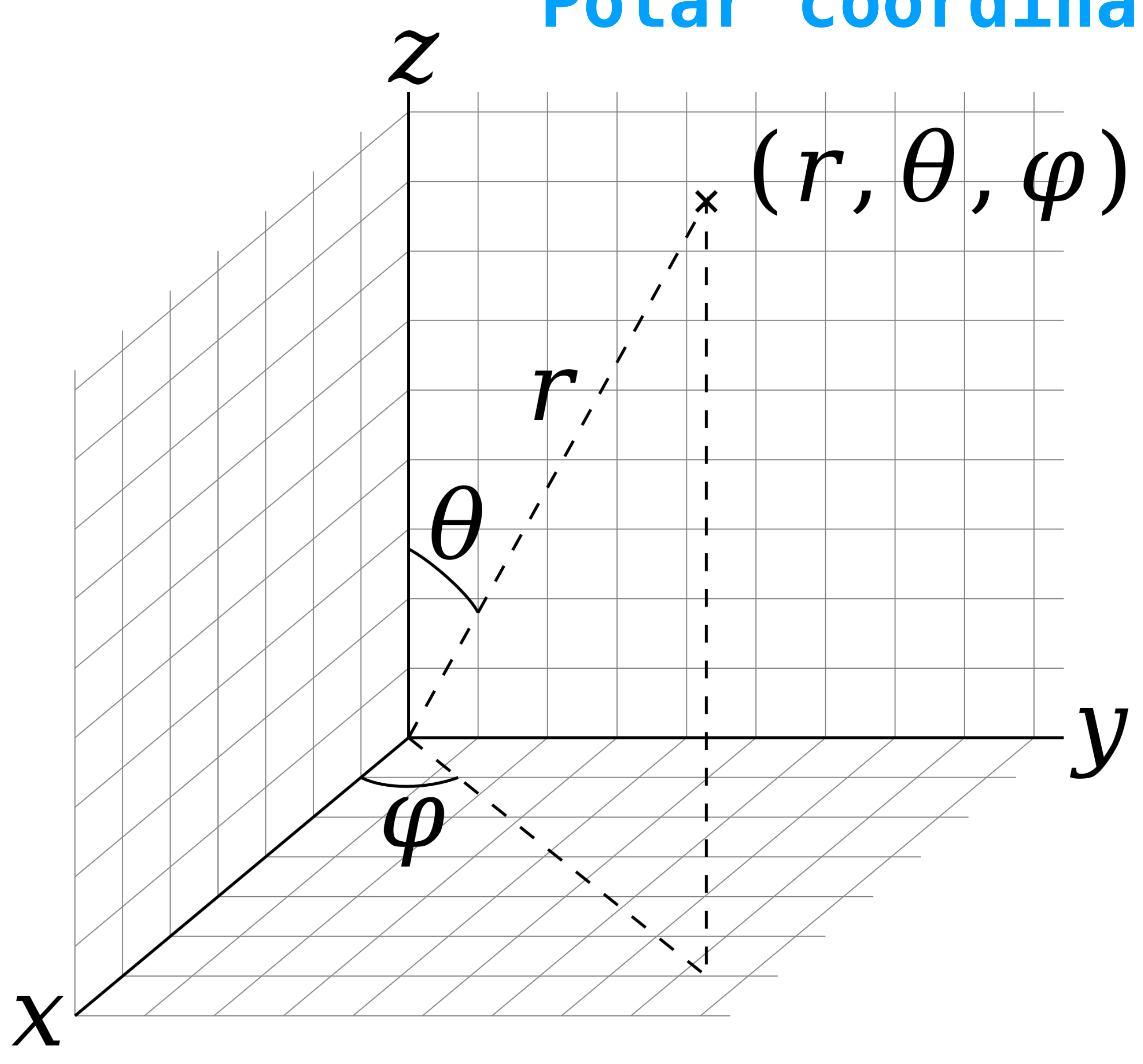
A coordinate system is a way of representing *positions* in terms of a *sequence of numbers*

Examples.

Cartesian coordinates



Polar coordinates



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It's just a sequence of numbers. We need to be *told* if it should be interpreted in the **polar** coordinate system or the **Cartesian** coordinate system

*Admittedly a trick question

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Given a basis $\mathscr{B}$ of a subspace H, there is **exactly one way** to write every vector in H as a linear combination of vectors in $\mathscr{B}$

Every basis provides a way to write down *coordinates* of a vector

And every time we write down a vector, we are **assuming a coordinate system**

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Then one day, you get tired of talking about "abstract" vectors, you want to work with *numbers*

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$$\mathbf{v} = 2\mathbf{b}_1 + 3\mathbf{b}_2 + \ldots + (-0.1)\mathbf{b}_n$$

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$$\mathbf{v} = 2\mathbf{b}_1 + 3\mathbf{b}_2 + \ldots + (-0.1)\mathbf{b}_n \qquad \mathbf{v} = \begin{bmatrix} 2 \\ 3 \\ \vdots \\ -0.1 \end{bmatrix}$$

and then choose those weights as a representation of v

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Every basis defined a different coordinate system

Standard Basis

The standard basis defines the Cartesian coordinate system for $\mathbb{R}^n$

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \ldots + v_n\mathbf{e}_n$$

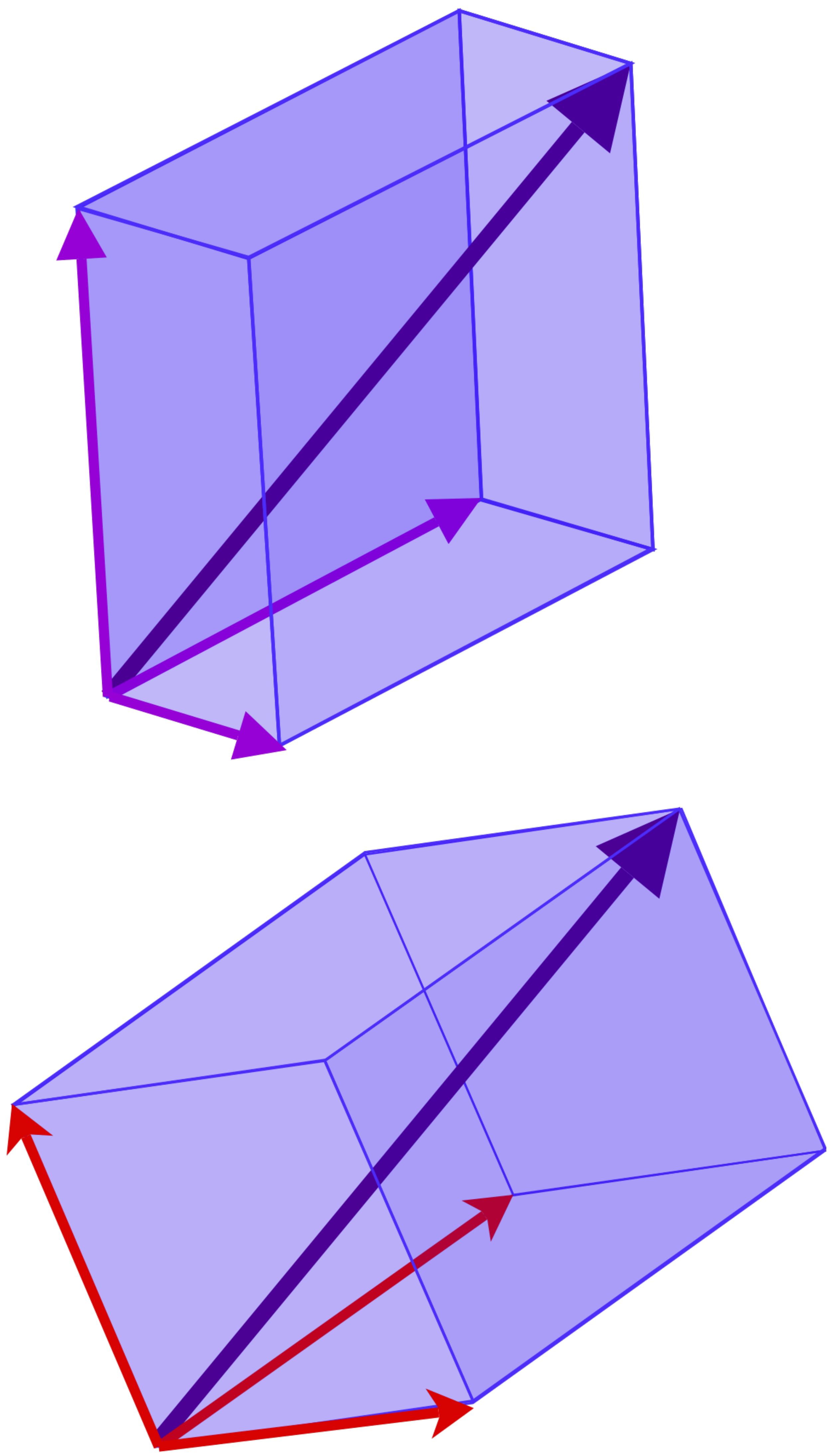
Column vectors are just weights for a linear combination of the standard basis

but we can also use
different coordinate systems

How to think about this

Changing the coordinate system "warps space"

The question is: how do we represent a vector $\mathbf{v}$ in the warped space if we wanted it to "be in the same place"?



https://commons.wikimedia.org/wiki/File:3d_two_bases_same_vector.svg

Coordinate Vectors

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Let $\mathbf{v}$ be a vector in a subspace H of $\mathbb{R}^n$ and let $\mathscr{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_k\}$ be a basis of H where

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Def. The **coordinate vector of $\mathbf{v}$ relative to $\mathscr{B}$** is

$$[\mathbf{v}]_{\mathscr{B}} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_k \end{bmatrix}$$

Coordinate Vectors and the Standard Basis

When we write down a vector $\mathbf{v}$ in $\mathbb{R}^n$, we're really writing down a coordinate vector **relative to the standard basis** $\mathscr{E}$

$$[\mathbf{v}]_{\mathscr{E}} = \mathbf{v}$$

How do we find coordinate vectors?

For an arbitrary basis $\mathscr{B}$, to determine $[\mathbf{v}]_{\mathscr{B}}$, we need to find weights $a_1, \ldots, a_k$ such that

$$a_1\mathbf{b}_1 + \ldots + a_k\mathbf{b}_k = \mathbf{v}$$

This is just *solving a vector equation*

How To: Coordinate Vectors

Q. Find the coordinate vector for $\mathbf{v}$ in the subspace H relative to the basis $\mathbf{b}_1, \dots, \mathbf{b}_k$

A. Solve the vector equation $x_1\mathbf{b}_1 + \dots + x_k\mathbf{b}_k = \mathbf{v}$. A solution $(a_1, \dots, a_k)$ means

$$[\mathbf{v}]_{\mathscr{B}} = \begin{bmatrix} a_1 \\ \vdots \\ a_k \end{bmatrix}$$

Dimension and Rank

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This number is a measure of how "large" H is

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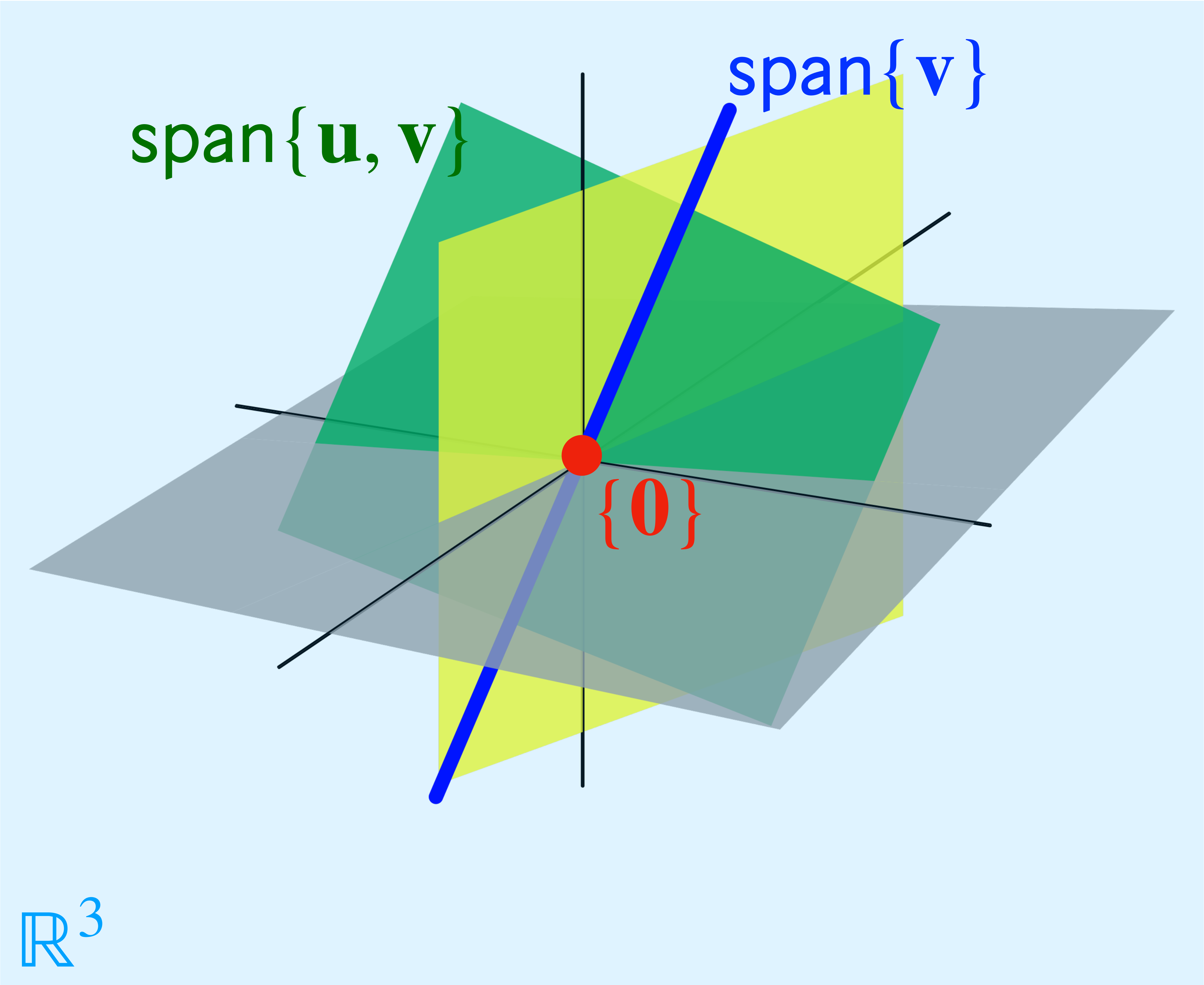
» a plane (through the origin) is a 2D subspace
» a line (through the origin) is a 1D subspace

Recall: Subspace in $\mathbb{R}^3$

There are only 4 kinds of subspaces of $\mathbb{R}^3$:

1. $\{\mathbf{0}\}$ just the origin
2. lines (through the origin)
3. planes (through the origin)
4. All of $\mathbb{R}^3$

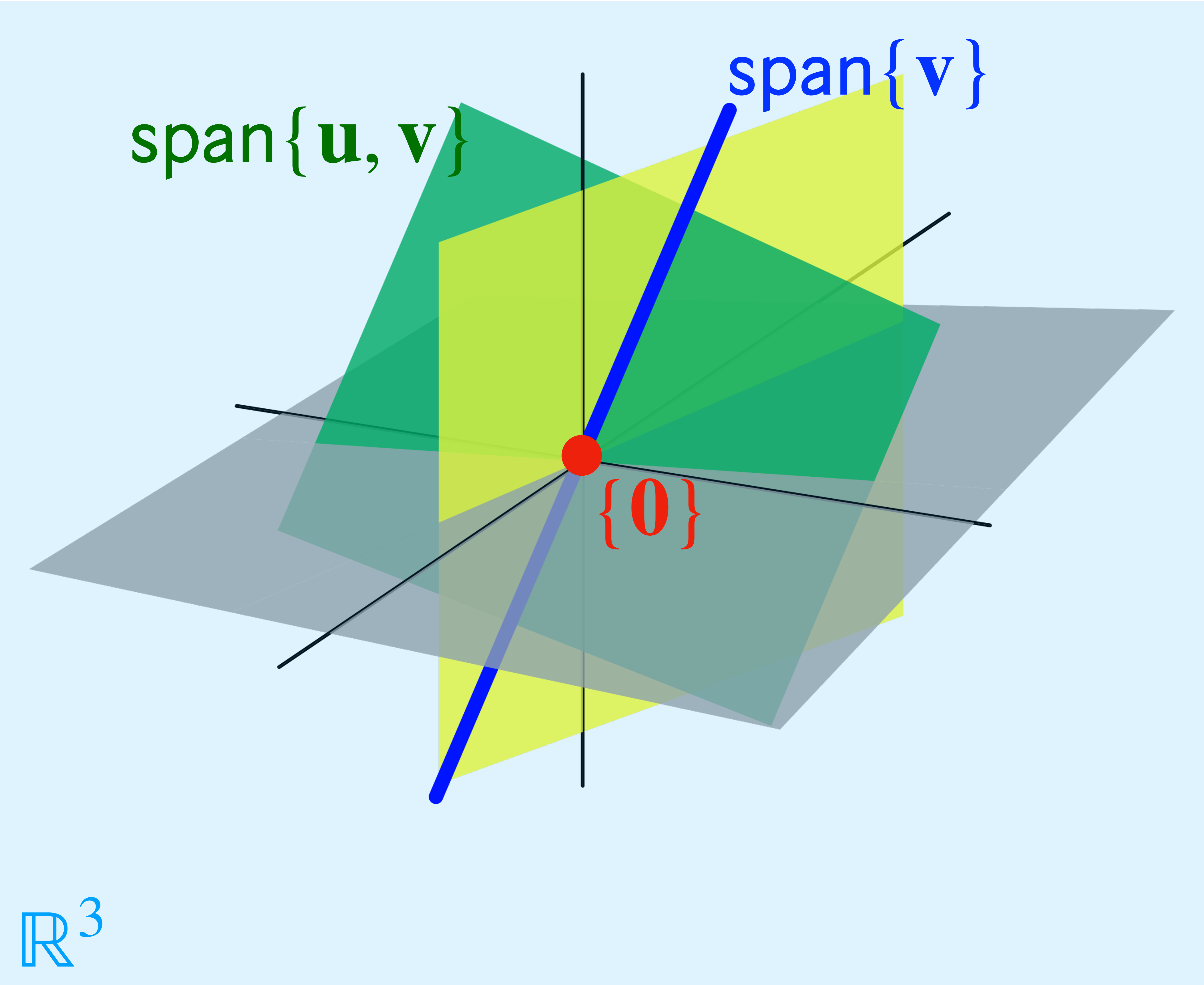


https://commons.wikimedia.org/wiki/File:Linear_subspaces_with_shading.svg

Recall: Subspace in $\mathbb{R}^3$

There are only 4 kinds of subspaces of $\mathbb{R}^3$:

1. 0-dimensional subspace
2. 1-dimensional subspace
3. 2-dimensional subspace
4. 3-dimensional subspace



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How does this connect to null space and column space?

Recall: An Observation

The *number* of vectors in the basis we found is the same as the number of *free variables* in a general form solution

$x_1 = 2x_2 + x_4 - 3x_5$

x_2 is free

$x_3 = (-2)x_4 + 2x_5$

x_4 is free

x_5 is free

$$\equiv \quad \begin{bmatrix} s \\ t \\ u \end{bmatrix} \mapsto \begin{bmatrix} 2s + t - 3u \\ s \\ (-2)t + 2u \\ t \\ u \end{bmatrix}$$

Dimension of the Null Space

The **dimension of** $\mathrm{Nul}(A)$ is the number of *free variables* in a general form solution to $A\mathbf{x} = \mathbf{0}$

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The *number* of vectors in the basis we found is the same as the number of *basic variable* or equivalently the number of *pivot columns*

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \mathbf{a}_3 \quad \mathbf{a}_4 \quad \mathbf{a}_5] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

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Dimension of the Column Space

The **dimension of** $\mathrm{Col}(A)$ is the number of *basic variable* in our solution, or equivalently the number of *pivot columns* of A

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Rank

Definition. The **rank** of a matrix A, written $\operatorname{rank} A$, is the dimension of $\operatorname{Col} A$

This is just terminology

Rank-Nullity Theorem

Theorem. For $A \in \mathbb{R}^{m \times n}$, $\operatorname{rank}(A) + \underbrace{\dim(\operatorname{Nul}(A))}_{\text{nullity}} = n$

This is incredibly important

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The null space "takes away" dimension from the column space

Extending the IMT

Theorem. For an $n \times n$ invertible matrix A, the following are logically equivalent (they must all by true or all by false

- $\mathrm{Col}(A) = \mathbb{R}^n$
- $\dim(\mathrm{Col}(A)) = n$
- $\mathrm{rank}(A) = n$
- $\mathrm{Nul}(A) = \{\mathbf{0}\}$
- $\dim(\mathrm{Nul}(A)) = 0$

workshop